Uniform quotient lower-bound construction

2026-11-10. Reviewed symbolic constraints; no experimental run.

Necessary size bound only: existence of a growing-degree quotient remains open

Failed block ansatz	Tensor-factorized x/y images imply actual solutions
Failed operator ansatz	Global commutation alone forces a Boolean solution representation
Necessary size	Actual quotient rank q is at least 2^(D-1)
Remaining construction	Coupled images, truncated boundary defects and full kernel closure

The exact Semaev coefficient tensor must annihilate the proposed quotient columns. Every lower-degree kernel relation must remain in the kernel after every legal variable multiplication. This includes feedback from newly derived consequences; an initial-matrix annihilator does not suffice.

The tensor obstruction follows already from block-bilinear input containment: normalized dual functionals on the two factors give actual Boolean assignments satisfying every generator. The operator obstruction needs only global commutation: idempotence on the cyclic vector then propagates to the entire spanned quotient.

The exponential size lemma uses a degree-compatible multiplicative monomial order. Leading kernel monomials are upward closed under squarefree multiplication within the cap. Standard monomials form a downset. An early filtration plateau would create a global solution module, so a useful separator retains a standard monomial of degree D-1 and all its2^(D-1) divisors.

The same divisor argument gives $\dim(Q_e) - \dim(Q_{e-1}) \geq \binom{D-1}{e}$ for $0 \leq e \leq D-1$, with $Q_{-1} = 0$. This constrains every layer of a proposed uniform filtration.

This bound concerns certificate representation size, not existence or LFD growth. Implicit exponential-dimensional constructions remain possible; polynomial rank in field size can still support logarithmic degree growth. A valid quotient must keep nonzero commutation defects at the truncated boundary, outside the lower-degree subspace where Boolean identities hold.

No explicit infinite growing-degree family was constructed. The next obligation is a coupled symbolic quotient for actual non-subfield factor bases, with a proof of full closure and a separate missing-consequence bridge. No new experiment, replay, build or cloud job ran.

Evidence: uniform-quotient-algebraic-20261110.md and review; uniform-quotient-operator-20261110.md and review; quotient-exponential-rank-review.md. These are written derivations, not formal-kernel certification.